SEPARABILITY IS STABLE IN TOWERS OF ALGEBRAIC EXTENSIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: This lemma is the permanence half behind the separable subextension E_sep used throughout the inseparability theory.

Editorial note. The label lemma-separable-permanence names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: lemma-separable-goes-up (separability of the top step propagates down the tower), lemma-separably-generated-separable (a finitely generated tower with separable steps is separable); irreducibility of the minimal polynomial over the bigger field E

Source reference. Original reference 09H2: sources/stacks-fields/pinned-original-source.tex, source lines 1404–1404. Prerequisite pointer invoked at 1641; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 030X: sources/stacks-fields/pinned-original-source.tex, source lines 1560–1560. Prerequisite pointer invoked at 1647; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let E/k and F/E be separable algebraic extensions of fields. Then F/k is a separable extension of fields.*

Proof. Choose $\alpha \in F$. Then α is separable algebraic over E. Let $P = x^d + \sum_{i<d} a_i x^i$ be the minimal polynomial of α over E. Each a_i is separable algebraic over k. Consider the tower of fields

$$k \subset k(a_0) \subset k(a_0, a_1) \subset \ldots \subset k(a_0, \ldots, a_{d-1}) \subset k(a_0, \ldots, a_{d-1}, \alpha)$$

Because a_i is separable algebraic over k it is separable algebraic over $k(a_0, \ldots, a_{i-1})$ by Lemma 09H2. Finally, α is separable algebraic over $k(a_0, \ldots, a_{d-1})$ because it is a root of P which is irreducible (as it is irreducible over the possibly bigger field E) and separable (as it is separable over E). Thus $k(a_0, \ldots, a_{d-1}, \alpha)$ is separable over k by Lemma 030X and we conclude that α is separable over k as desired. □

Date: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.